

Exponential Functions & Growth Models

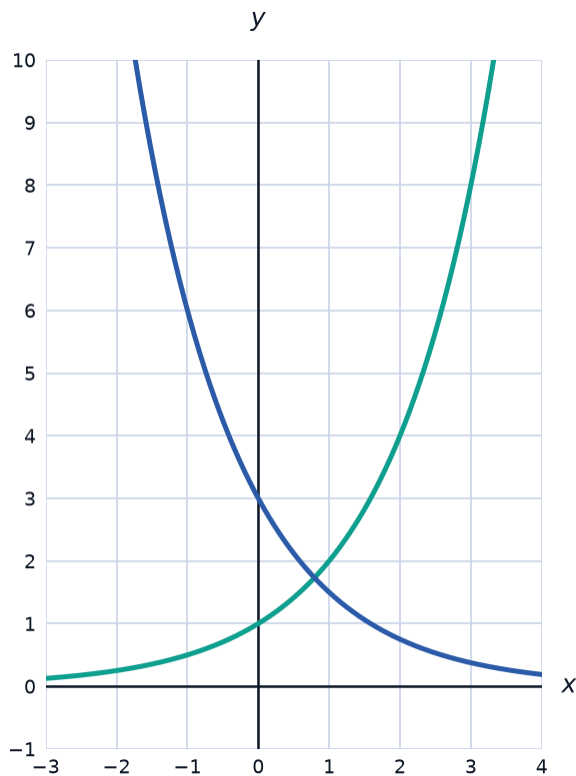


Exponential growth and decay basics

- 1 Classify each function as exponential growth or decay, and state the growth/decay factor: $f(x) = 5(1.2)^x$ and $g(x) = 5(0.8)^x$.
- 2 A population of bacteria is modeled by $P(t) = 1000(2)^{t/3}$, where t is in hours. Interpret the meaning of the exponent's structure, and find the population after 9 hours.

Visualizing exponential growth

- 3 The graph shows $f(x) = 2^x$ and $g(x) = 3(0.5)^x$, one growing and one decaying.



- a Identify which curve represents growth and which represents decay.
- b State the horizontal asymptote shared by both curves.

Finding an exponential model from data

- 4 An investment grows from 2000 dollars to 2420 dollars in 2 years under exponential growth $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

- 5 A radioactive sample decays from 80 g to 45 g over 6 years. Find the annual decay factor b in the model $A(t) = 80b^t$, to four decimal places.
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Continuous growth and the number e

- 6 A quantity grows continuously according to $A(t) = 500e^{0.03t}$. Find the amount after 20 years, to the nearest whole number.
- 7 Explain the difference between the discrete growth model $A(t) = A_0(1 + r)^t$ and the continuous growth model $A(t) = A_0e^{kt}$, and state how r and k are related when both models describe the same annual growth rate r .
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Comparing growth rates: exponential vs. polynomial

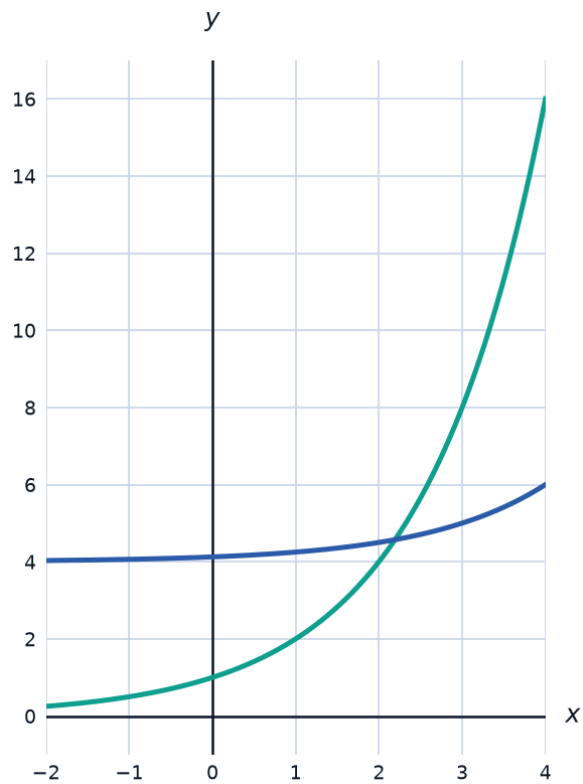
- 8 Explain why, for large enough x , any exponential growth function $f(x) = b^x$ (with $b > 1$) will eventually exceed any polynomial function $g(x) = x^n$, no matter how large the exponent n is. Illustrate with $f(x) = 1.01^x$ vs. $g(x) = x^{10}$.
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Applications: half-life and doubling time

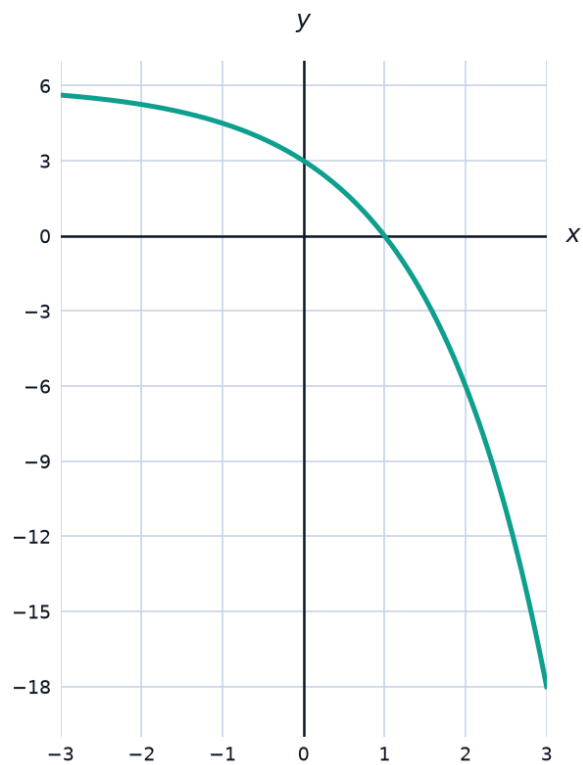
- 9 A substance has a half-life of 8 days. Write its decay model $A(t) = A_0\left(\frac{1}{2}\right)^{t/8}$, then find what fraction of the original amount remains after 20 days.
- 10 A city's population doubles every 15 years and currently stands at 40,000. Write an exponential model $P(t)$ and predict the population after 45 years.
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Transformations of exponential functions

- 11 Describe how the graph of $g(x) = 2^{x-3} + 4$ is transformed from the parent function $f(x) = 2^x$, and state the new horizontal asymptote.



- 12 Find the horizontal asymptote and y-intercept of $h(x) = -3(2)^x + 6$.



Compound interest and financial models

- 13** A savings account earns 4% annual interest compounded quarterly, modeled by $A(t) = A_0 \left(1 + \frac{0.04}{4}\right)^{4t}$. If 5000 dollars is deposited, find the balance after 3 years, to the nearest dollar.
- 14** Explain, without computing exact values, why compounding more frequently (e.g. daily instead of quarterly) at the same nominal annual rate produces a slightly larger balance after the same number of years.
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Multiple choice — easy 15

- 15** A population is modeled by $P(t) = 900(3)^{t/3}$, where t is in years. Find $P(6)$.
- a. 2700
 - b. 5400
 - c. 8100
 - d. 24300
- 16** A population is modeled by $P(t) = 200(3)^{t/6}$, where t is in years. Find $P(6)$.
- a. 800
 - b. 600
 - c. 400
 - d. 1800
- 17** A population is modeled by $P(t) = 700(3)^{t/3}$, where t is in years. Find $P(3)$.
- a. 2800
 - b. 6300
 - c. 1400
 - d. 2100

- 18 A population is modeled by $P(t) = 400(2)^{t/7}$, where t is in years. Find $P(7)$.
- a. 1200
 - b. 1600
 - c. 800
 - d. 400
- 19 A population is modeled by $P(t) = 700(2)^{t/2}$, where t is in years. Find $P(4)$.
- a. 5600
 - b. 1400
 - c. 2100
 - d. 2800
- 20 A population is modeled by $P(t) = 700(2)^{t/2}$, where t is in years. Find $P(6)$.
- a. 11200
 - b. 4200
 - c. 2800
 - d. 5600
- 21 A population is modeled by $P(t) = 500(3)^{t/3}$, where t is in years. Find $P(3)$.
- a. 1000
 - b. 2500
 - c. 4500
 - d. 1500

- 22 A population is modeled by $P(t) = 500(2)^{t/4}$, where t is in years. Find $P(16)$.
- a. 9000
 - b. 8000
 - c. 16000
 - d. 4000
- 23 A population is modeled by $P(t) = 100(2)^{t/4}$, where t is in years. Find $P(8)$.
- a. 800
 - b. 400
 - c. 200
 - d. 300
- 24 A population is modeled by $P(t) = 900(3)^{t/5}$, where t is in years. Find $P(15)$.
- a. 26100
 - b. 8100
 - c. 24300
 - d. 72900
- 25 A population is modeled by $P(t) = 700(3)^{t/2}$, where t is in years. Find $P(6)$.
- a. 20300
 - b. 56700
 - c. 18900
 - d. 6300

- 26 A population is modeled by $P(t) = 200(2)^{t/5}$, where t is in years. Find $P(20)$.
- a. 3200
 - b. 6400
 - c. 1600
 - d. 3600
- 27 A population is modeled by $P(t) = 800(3)^{t/2}$, where t is in years. Find $P(2)$.
- a. 4000
 - b. 2400
 - c. 3200
 - d. 7200
- 28 A population is modeled by $P(t) = 500(2)^{t/5}$, where t is in years. Find $P(20)$.
- a. 8500
 - b. 8000
 - c. 4000
 - d. 16000
- 29 A population is modeled by $P(t) = 600(2)^{t/6}$, where t is in years. Find $P(12)$.
- a. 4800
 - b. 2400
 - c. 1200
 - d. 1800

Multiple choice — medium 15

30 A substance has a half-life of 6 days. What fraction of the original amount remains after 18 days?

a. $\frac{1}{4}$

b. $\frac{1}{16}$

c. $\frac{3}{8}$

d. $\frac{1}{8}$

31 A substance has a half-life of 6 days. What fraction of the original amount remains after 30 days?

a. $\frac{1}{16}$

b. $\frac{5}{32}$

c. $\frac{1}{32}$

d. $\frac{1}{64}$

32 A substance has a half-life of 2 days. What fraction of the original amount remains after 2 days?

a. $\frac{1}{4}$

b. $\frac{5}{8}$

c. $\frac{1}{2}$

d. $\frac{3}{4}$

33 A substance has a half-life of 5 days. What fraction of the original amount remains after 15 days?

a. $\frac{1}{16}$

b. $\frac{1}{4}$

c. $\frac{1}{8}$

d. $\frac{3}{8}$

- 34 A substance has a half-life of 2 days. What fraction of the original amount remains after 6 days?
- a. $\frac{1}{4}$
 - b. $\frac{1}{8}$
 - c. $\frac{1}{16}$
 - d. $\frac{3}{8}$
- 35 A substance has a half-life of 6 days. What fraction of the original amount remains after 12 days?
- a. $\frac{5}{16}$
 - b. $\frac{1}{8}$
 - c. $\frac{1}{4}$
 - d. $\frac{1}{2}$
- 36 A substance has a half-life of 8 days. What fraction of the original amount remains after 24 days?
- a. $\frac{1}{16}$
 - b. $\frac{1}{4}$
 - c. $\frac{3}{8}$
 - d. $\frac{1}{8}$
- 37 A substance has a half-life of 7 days. What fraction of the original amount remains after 21 days?
- a. $\frac{1}{8}$
 - b. $\frac{3}{8}$
 - c. $\frac{1}{16}$
 - d. $\frac{1}{4}$

- 38 A substance has a half-life of 7 days. What fraction of the original amount remains after 35 days?
- a. $\frac{1}{32}$
 - b. $\frac{5}{32}$
 - c. $\frac{1}{64}$
 - d. $\frac{1}{16}$
- 39 A substance has a half-life of 3 days. What fraction of the original amount remains after 9 days?
- a. $\frac{1}{8}$
 - b. $\frac{3}{8}$
 - c. $\frac{1}{4}$
 - d. $\frac{1}{16}$
- 40 A substance has a half-life of 4 days. What fraction of the original amount remains after 4 days?
- a. $\frac{5}{8}$
 - b. 1
 - c. $\frac{1}{4}$
 - d. $\frac{1}{2}$
- 41 A substance has a half-life of 2 days. What fraction of the original amount remains after 10 days?
- a. $\frac{5}{32}$
 - b. $\frac{1}{64}$
 - c. $\frac{1}{16}$
 - d. $\frac{1}{32}$

42 A substance has a half-life of 5 days. What fraction of the original amount remains after 25 days?

a. $\frac{5}{32}$

b. $\frac{1}{64}$

c. $\frac{1}{16}$

d. $\frac{1}{32}$

43 A substance has a half-life of 4 days. What fraction of the original amount remains after 12 days?

a. $\frac{1}{8}$

b. $\frac{1}{16}$

c. $\frac{1}{4}$

d. $\frac{3}{8}$

44 A substance has a half-life of 9 days. What fraction of the original amount remains after 45 days?

a. $\frac{1}{64}$

b. $\frac{1}{16}$

c. $\frac{1}{32}$

d. $\frac{5}{32}$

Multiple choice — hard 15

45 An investment grows from 300 dollars to 507 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .

a. $\frac{3}{10}$

b. $\frac{2}{5}$

c. $-\frac{3}{10}$

d. $\frac{13}{10}$

- 46 An investment grows from 200 dollars to 128 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .
- a. $-\frac{1}{10}$
 - b. $-\frac{1}{5}$
 - c. $\frac{1}{5}$
 - d. $\frac{4}{5}$
- 47 An investment grows from 100 dollars to 169 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .
- a. $-\frac{3}{10}$
 - b. $\frac{3}{10}$
 - c. $\frac{2}{5}$
 - d. $\frac{13}{10}$
- 48 An investment grows from 200 dollars to 98 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .
- a. $-\frac{3}{10}$
 - b. $\frac{3}{10}$
 - c. $\frac{7}{10}$
 - d. $-\frac{1}{5}$
- 49 An investment grows from 200 dollars to 162 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r .
- a. $\frac{9}{10}$
 - b. $\frac{1}{10}$
 - c. 0
 - d. $-\frac{1}{10}$

- 50 Find the horizontal asymptote and y -intercept of $h(x) = -7(2)^x - 3$.
- a. Horizontal asymptote $y = -10$; y -intercept (-3) .
 - b. Horizontal asymptote $y = -3$; y -intercept (-7) .
 - c. Horizontal asymptote $y = -3$; y -intercept (-10) .
 - d. Horizontal asymptote $y = 0$; y -intercept (-10) .
- 51 Find the horizontal asymptote and y -intercept of $h(x) = 5(2)^x + 1$.
- a. Horizontal asymptote $y = 6$; y -intercept (1) .
 - b. Horizontal asymptote $y = 1$; y -intercept (6) .
 - c. Horizontal asymptote $y = 1$; y -intercept (5) .
 - d. Horizontal asymptote $y = 0$; y -intercept (6) .
- 52 Find the horizontal asymptote and y -intercept of $h(x) = -3(2)^x - 7$.
- a. Horizontal asymptote $y = -7$; y -intercept (-3) .
 - b. Horizontal asymptote $y = -7$; y -intercept (-10) .
 - c. Horizontal asymptote $y = 0$; y -intercept (-10) .
 - d. Horizontal asymptote $y = -10$; y -intercept (-7) .
- 53 Find the horizontal asymptote and y -intercept of $h(x) = 3(2)^x - 4$.
- a. Horizontal asymptote $y = -1$; y -intercept (-4) .
 - b. Horizontal asymptote $y = -4$; y -intercept (-1) .
 - c. Horizontal asymptote $y = -4$; y -intercept (3) .
 - d. Horizontal asymptote $y = 0$; y -intercept (-1) .

- 54 Find the horizontal asymptote and y-intercept of $h(x) = 7(2)^x + 5$.
- a. Horizontal asymptote $y = 12$; y-intercept (5).
 - b. Horizontal asymptote $y = 5$; y-intercept (12).
 - c. Horizontal asymptote $y = 0$; y-intercept (12).
 - d. Horizontal asymptote $y = 5$; y-intercept (7).
- 55 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{20})^{nt}$, where $nt = 15$. If 2000 dollars is deposited, find the balance to the nearest dollar.
- a. 4008
 - b. 4258
 - c. 2000
 - d. 4158
- 56 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 9$. If 5000 dollars is deposited, find the balance to the nearest dollar.
- a. 7167
 - b. 7117
 - c. 5000
 - d. 7067
- 57 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{3}{100})^{nt}$, where $nt = 6$. If 5000 dollars is deposited, find the balance to the nearest dollar.
- a. 5970
 - b. 6120
 - c. 5920
 - d. 5000

- 58 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{25})^{nt}$, where $nt = 13$. If 1000 dollars is deposited, find the balance to the nearest dollar.
- a. 1000
 - b. 1515
 - c. 1665
 - d. 1765
- 59 A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{100})^{nt}$, where $nt = 16$. If 4000 dollars is deposited, find the balance to the nearest dollar.
- a. 4000
 - b. 4590
 - c. 4740
 - d. 4690